



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CURASCAN: INTELLIGENT HEALTHCARE IMAGING

The domain of the Project: Deep Learning

COURSE NAME

G37 AI & ML

Team Mentors (and their designation):

Gaurav Patel – Data Engineer (Celebal Technologies)

Team Members:

Ms. Nishitha Sharanu

Period of the project

2 Months

April 2026 to June 2026



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Declaration

The project titled “CURASCAN – Intelligent Healthcare Imaging” has been mentored by Gaurav Patel, organised by SURE Trust, from Nov 2025 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. ---S.Nishitha----- Signature

Mentor's Name – Gaurav Patel

Designation— Data Engineer

Company Name - Celebal Technologies

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

CURASCAN: Intelligent Healthcare Imaging is an AI-powered medical imaging system developed to assist healthcare professionals in the faster and more accurate diagnosis of diseases from chest X-rays, CT scans, and MRI images. The primary objective of the project was to reduce radiologists' workload, minimize diagnostic errors, and improve healthcare accessibility through automated medical image analysis.

The system utilizes **DenseNet-121** for disease classification, **U-Net** for segmentation of abnormal regions, and **Grad-CAM** for generating explainable visual heatmaps that highlight the areas influencing AI predictions. A user-friendly **Streamlit-based dashboard** was developed to enable image upload, analysis, visualization, and report generation.

Testing results demonstrated that the system could successfully distinguish between normal and abnormal medical images while accurately highlighting affected regions and providing visual explanations for predictions. This improves transparency and supports clinicians in making informed diagnostic decisions.

The project concludes that AI-assisted medical imaging can significantly enhance diagnostic efficiency, reduce human error, and support early disease detection. Future enhancements include expanding support for additional imaging modalities, improving accuracy with larger datasets, enabling real-time processing, integrating with hospital systems such as PACS and EMR, and developing mobile or offline versions for deployment in resource-limited healthcare environments.

Key Recommendations:

- Expand training datasets to improve model robustness and reduce bias.
- Integrate the system with existing hospital infrastructure.
- Enhance explainability features to increase clinician trust.
- Develop lightweight and mobile-friendly deployments for rural healthcare centers.
- Implement continuous learning mechanisms using clinician feedback.



Introduction

Background and Context of the Project

Medical imaging plays a vital role in modern healthcare by enabling doctors to diagnose and monitor diseases using techniques such as Chest X-rays, Computed Tomography (CT), and Magnetic Resonance Imaging (MRI). With the increasing number of patients and medical scans, radiologists often face heavy workloads, which can lead to delays in diagnosis and increase the risk of human error. Recent advancements in Artificial Intelligence (AI) and Deep Learning have created opportunities to automate image analysis and support healthcare professionals in making faster and more accurate clinical decisions. CURASCAN: Intelligent Healthcare Imaging was developed to leverage these technologies and enhance the efficiency of medical image interpretation.

Problem Statement and Goals of the Project

The growing demand for medical imaging services and the limited availability of skilled radiologists have created challenges in delivering timely and accurate diagnoses. Manual analysis of CT, MRI, and X-ray scans is often time-consuming and susceptible to fatigue-related errors. The primary goal of CURASCAN is to develop an AI-powered healthcare imaging system capable of automatically detecting diseases, identifying abnormal regions, and providing visual explanations for its predictions. The system aims to assist radiologists by reducing diagnostic workload, improving accuracy, and supporting early disease detection.

Scope and Limitations of the Project

The scope of CURASCAN includes the analysis of Chest X-rays, CT scans, and MRI images using deep learning techniques. The system performs disease classification through DenseNet-121, abnormal region segmentation using U-Net, and explainable visualization through Grad-CAM heatmaps. It also provides an interactive Streamlit-based dashboard for image upload and result interpretation. However, the current implementation is limited by the availability and diversity of training datasets and is intended to function as a decision-support tool rather than a replacement for medical professionals. Clinical validation on large-scale real-world datasets is required before deployment in healthcare environments.



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Innovation Component in the Project

The key innovation of CURASCAN lies in the integration of multiple AI technologies into a unified healthcare imaging platform. The system combines DenseNet-121 for disease classification, U-Net for precise segmentation of abnormal regions, and Grad-CAM for explainable AI visualization. This approach not only provides accurate predictions but also offers visual evidence through heatmaps, improving transparency and clinician trust. The multi-modality support for X-rays, CT scans, and MRI images, along with an easy-to-use web interface, makes CURASCAN a practical and scalable solution for intelligent medical image analysis.



Project Objectives

Project Objectives

Clearly Defined Objectives and Goals of the Project

- To develop an AI-powered healthcare imaging system for analyzing Chest X-rays, CT scans, and MRI images.
- To assist radiologists and healthcare professionals in detecting diseases quickly and accurately.
- To automate the medical image analysis process and reduce manual diagnostic effort.
- To implement **DenseNet-121** for disease classification in medical images.
- To utilize **U-Net** for precise segmentation and localization of abnormal regions.
- To integrate **Grad-CAM** for generating visual heatmaps that explain AI predictions.
- To improve diagnostic accuracy and reduce the chances of human error.
- To provide a user-friendly interface for uploading, analyzing, and interpreting medical images.
- To support early disease detection and enhance clinical decision-making.
- To make quality healthcare diagnostics more accessible, especially in resource-limited environments.

Expected Outcomes and Deliverables

- A fully functional **CURASCAN** healthcare imaging application.
- An AI model capable of classifying medical images as normal or abnormal.
- A segmentation system that highlights affected regions in medical scans.
- Grad-CAM heatmap visualizations for explainable and transparent AI predictions.
- A Streamlit-based web dashboard for image upload, analysis, and result visualization.
- Automated generation of diagnostic insights and visual reports.
- Performance evaluation of the classification and segmentation models.
- Technical documentation, user manuals, and deployment guidelines.
- A comprehensive project report detailing methodology, implementation, results, and future scope.
- A scalable framework that can be further enhanced with larger datasets and additional imaging modalities.



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Methodology and Results

5.1 Methods / Technology Used

- Developed a web-based healthcare imaging system using **Python** and **Streamlit**.
- Implemented **Deep Learning** models for medical image analysis and disease prediction.
- Used **Convolutional Neural Networks (CNNs)** and **ResNet architecture** for image classification.
- Applied image preprocessing techniques such as resizing, normalization, and enhancement.
- Integrated database functionality for storing patient information, scan details, and prediction history.
- Generated automated diagnostic reports based on model predictions.
- Provided a user-friendly dashboard for doctors and healthcare professionals.

5.2 Tools / Software Used

Operating system: Windows 8 or later or Linux (Ubuntu 20.04 and above)

Programming Language: Python 3.9 or higher.

IDE/Code Editor: Visual Studio Code/ PyCharm.

Frameworks: PyTorch, Torchvision, Streamlit.

Additional Tools : Streamlit for web deployment.

Office/Documentation: MS Office / LibreOffice / Google Docs.

Version Control: Git + GitHub/GitLab/Bitbucket.

5.3 Data Collection Approach

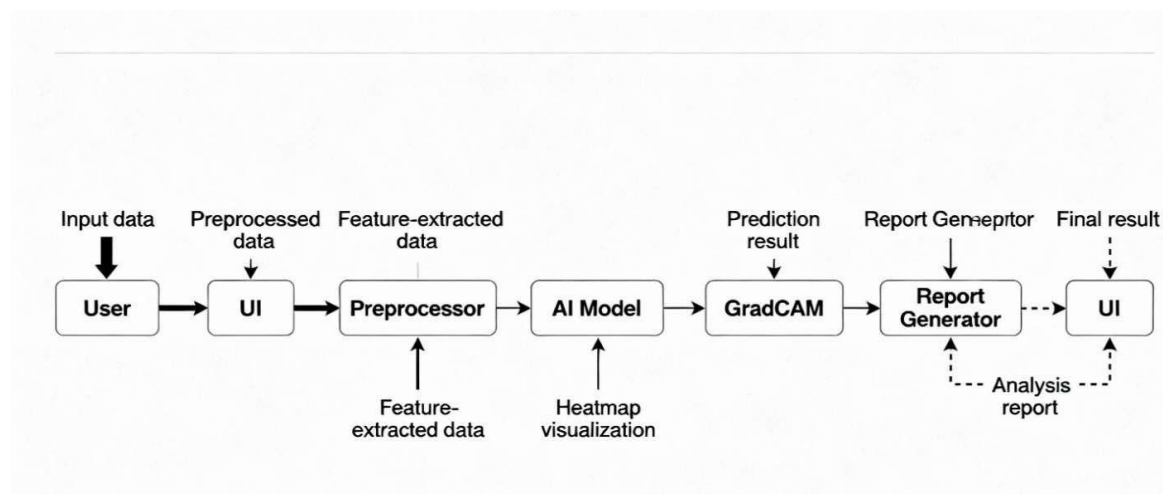
- Medical imaging datasets were collected from publicly available healthcare repositories.
- Images included various scan types such as:
 - X-Ray Images
 - MRI Scans
 - CT Scans
 - Skin Disease Images (if included)
- Data preprocessing steps:
 - Removal of corrupted images.



- Image resizing to standard dimensions.
- Pixel normalization.
- Data augmentation (rotation, flipping, zooming) to improve model performance.
- Dataset was divided into:
 - Training Set (70%)
 - Validation Set (15%)
 - Testing Set (15%)

5.4 Project Architecture

Architecture Flow



Architecture Description

1. User uploads a medical image through the Streamlit interface.
2. The system preprocesses the image by resizing and normalization.
3. The processed image is passed to the trained deep learning model.
4. The model predicts the disease category and confidence score.
5. Results are displayed to the user through the dashboard.
6. Prediction history and patient records are stored in the SQLite database.
7. Reports can be generated and reviewed by healthcare professionals.



5.5 Results

To evaluate the performance of the proposed system, a small set of medical images was tested using the developed models. In this test, four images were used as input samples. Among them, two images represented normal cases, while the other two contained abnormal medical conditions. The classification model based on DenseNet-121 analyzed these images and was able to correctly identify whether the scan was normal or abnormal. For the images classified as abnormal, the system further highlighted the possible affected areas using the U-Net model. This model generated segmentation masks that clearly marked the regions of interest in the medical scans, helping users understand where the abnormality might be located.

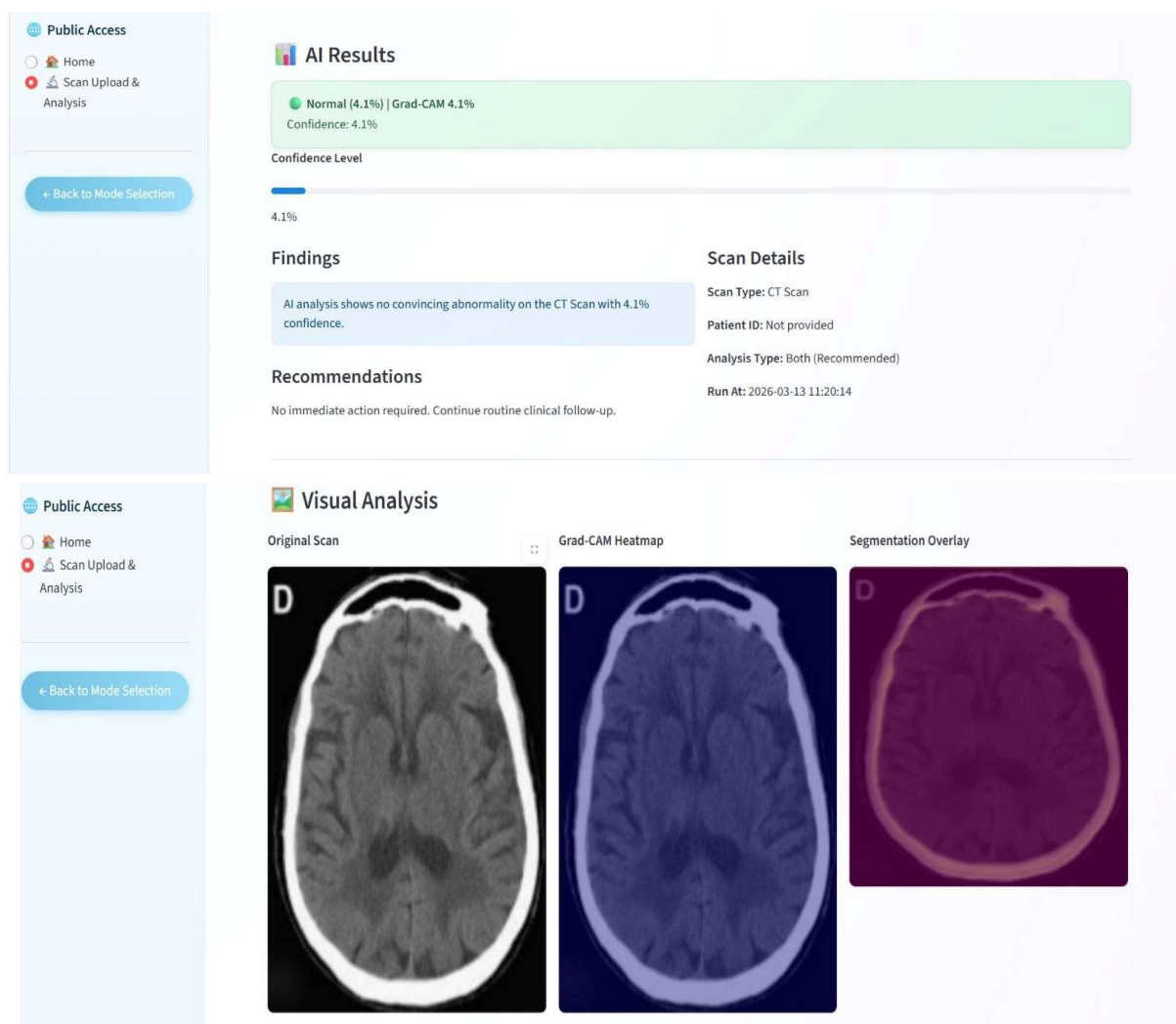


Fig: AI Analysis Result Showing Normal CT Scan Prediction Visual Analysis of CT Scan



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To make the predictions easier to understand, the system also uses Grad-CAM. This method creates heatmaps that show which parts of the image influenced the model's decision. These heatmaps are displayed on top of the original image so that users can visually see the areas that contributed to the prediction. All results are displayed through a simple dashboard developed using Streamlit. Users can upload a medical image, run the analysis, and view the prediction results along with segmentation masks and heatmaps. The test results show that the system can successfully distinguish between normal and abnormal images while also providing visual explanations for the predictions.

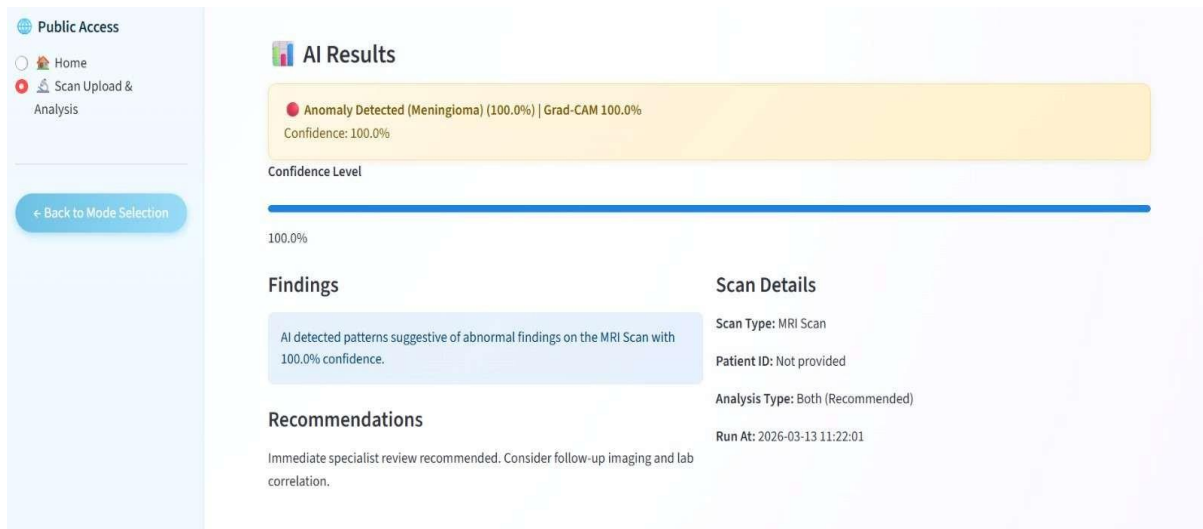


Fig: AI Detection Result Indicating Abnormal (Meningioma) Visual Explanation of MRI Scan

Key Achievements

- Successfully classified medical images using deep learning models.
- Generated accurate disease predictions with confidence scores.
- Provided an intuitive dashboard for healthcare practitioners.
- Maintained patient records and scan history.
- Enabled report generation for clinical review.

System Performance

- Fast image processing and prediction generation.
- Efficient database management for patient records.
- Responsive web-based user interface.
- Real-time visualization of diagnostic results.



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5.6 Project GitHub Link

GitHub Repository: <https://github.com/sure-trust/NISHITHA-SHARANU-g37-ai-ml/tree/7b534cc019744841e5f9c391b75cd52aaaef2df5/Final%20capstone%20project/CURASCAN-main>

Social / Industry Relevance of the Project

CURASCAN: Intelligent Healthcare Imaging has significant social and industrial relevance as it addresses the growing demand for fast, accurate, and accessible medical diagnosis. In the healthcare sector, radiologists often face heavy workloads due to the increasing number of medical scans, which can lead to delays in diagnosis and potential human errors. CURASCAN helps overcome these challenges by using Artificial Intelligence to automate medical image analysis, enabling quicker and more reliable detection of diseases.

From a social perspective, the project contributes to improving healthcare accessibility, especially in rural and under-resourced areas where experienced radiologists may not be readily available. By supporting early disease detection and reducing diagnostic delays, the system can help improve patient outcomes and overall quality of healthcare services.

From an industry perspective, the project demonstrates the practical application of Artificial Intelligence, Deep Learning, Computer Vision, and Explainable AI in healthcare. It aligns with the ongoing digital transformation of the medical industry and showcases how intelligent systems can enhance clinical workflows, reduce operational costs, improve efficiency, and support healthcare professionals in making informed decisions. The integration of explainable AI also increases trust and transparency, making the technology more suitable for real-world clinical adoption.



Learning and Reflection

New Learnings

Through the development of **CURASCAN: Intelligent Healthcare Imaging**, I gained practical knowledge in several technical and professional areas. I learned how to apply deep learning models such as DenseNet-121 for medical image classification and U-Net for image segmentation. I also gained hands-on experience with Explainable AI techniques, particularly Grad-CAM, to generate visual explanations for model predictions. The project enhanced my skills in Python, PyTorch, OpenCV, Streamlit, data preprocessing, model training, testing, and deployment. Additionally, I learned how AI can be applied in healthcare to support disease detection and medical image analysis.

Apart from technical skills, I improved my problem-solving abilities, project planning, debugging skills, documentation practices, and teamwork. I also learned how to manage project timelines, coordinate with team members, and integrate different modules into a complete working system.

Overall Experience

Working on CURASCAN was a valuable and enriching experience. It provided me with an opportunity to work on a real-world healthcare application and understand the complete machine learning project lifecycle, from data collection and preprocessing to model development, deployment, and evaluation. Throughout the project, I faced challenges related to data handling, model optimization, and system integration, which helped me develop patience and analytical thinking. The experience strengthened my confidence in developing AI-based applications and deepened my interest in machine learning, computer vision, and healthcare technology. Overall, the project was both technically rewarding and professionally beneficial, helping me bridge the gap between academic concepts and practical implementation.



Conclusion and Future Scope

In this project, we developed CURASCAN, an intelligent medical imaging system designed to support doctors in detecting abnormalities across multiple imaging modalities such as chest X-rays, CT scans, and MRI more efficiently and accurately. The core motivation behind this work was to demonstrate how AI and deep learning can assist healthcare professionals, especially in situations where radiologists face heavy workloads or where specialists are not readily available. By integrating advanced models like DenseNet121 for classification and U-Net for segmentation, the system can analyze medical images, identify potential problem areas, and visually highlight them for easier interpretation. The use of Grad-CAM heatmaps adds transparency by helping clinicians understand the reasoning behind predictions, thereby improving trust and decision-making. Although still evolving, this project highlights the strong potential of AI in clinical environments, and with further improvements, CURASCAN can become a powerful tool in hospitals and diagnostic centers, supporting a future where AI and human expertise work together to deliver better healthcare outcomes.

To further enhance CURASCAN for medical applications, several improvements are planned, including expanding support for more image types such as CT scans, MRI images, and ultrasound beyond chest Xrays, and improving prediction accuracy by using larger and more diverse datasets from multiple hospitals to reduce bias. The system can also be optimized for faster processing to deliver real-time results, which is crucial in emergency situations, while improving model transparency by clearly highlighting the regions that influence diagnostic decisions to build greater clinical trust. Additionally, integrating CURASCAN with hospital systems like PACS or EMR will streamline workflows, and developing a lightweight mobile or offline version will enable quick screenings in rural or resource-limited areas. Incorporating a continuous learning mechanism, where doctors can provide feedback and correct predictions, will further enhance the system's performance over time.